Sealed by Default: Privacy-Preserving Parked-Vehicle Camera Recording with Fail-Closed Sealing and Owner-or-Court Unsealing

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Abstract

Parked vehicles increasingly record every passer-by, while European regulators hold owners responsible for footage they cannot design. Raw recording keeps evidence of damage but exposes every face and number plate; irreversible blurring protects bystanders but destroys the evidence. We propose a sealed-by-default architecture in which identity regions are separated inside the camera pipeline, stored as exactly recoverable encrypted pixels under per-person keys, and released only for a specific incident with the owner's key or a court order; the head region of any person without a detected face is sealed as well (fail-closed). We prove that sealing never exposes more than raw recording, retains the same evidence under exact recovery, and, under two stated assumptions, bounds residual exposure by the person-detector miss rate at the identification resolution. In a geometric simulator with ground-truth identifiability at the IEC 62676-4 identification level (810 paired episodes), the proposed design reduced owner-accessible identifiable exposure from 59.5 s (10.5 s of it faces) to 0.0 s per 10 min relative to event-triggered raw recording, matched its evidence utility (0.52), and exceeded fail-closed blurring by 0.52 in evidence utility at equal exposure. Per-person keys avoided 1.08 s of bystander exposure per unseal. Fail-closed protection held until person detection required the identification resolution itself. We also map the design to EU law as of 2026 and propose eight regulatory amendments. Absolute magnitudes depend strongly on assumed trigger, detector and condition models, while the relative conclusions held in every sensitivity run; real-vehicle validation is required.

Index Terms— *Automotive cameras, bystander privacy, GDPR, anonymization, key escrow, parked-vehicle surveillance, regulation, sentry mode, simulation.*

I Introduction

A parked car with surround cameras is a privately operated surveillance unit on public ground. Sentry-type functions record people who walk past without notice or consent, and the footage is stored where owners, manufacturer staff, police or attackers may access it. European data protection authorities have so far placed controller duties on the vehicle owner rather than the manufacturer [1], although the household exemption does not cover cameras that capture public space [2]. An Austrian court held that live capture by such cameras is processing even when nothing is stored [3], and a Belgian minister declared continuous use of a widely sold sentry function incompatible with national camera law [4]. Footage has been shared internally by manufacturer staff [5], requested by police by towing parked vehicles [6], and exposed in a breach of a dashcam image service [7]. A German state regulator reported fining drivers and not having seen a mobile dashcam that can be operated lawfully [8][RSV]. China instead mandates default anonymization of exterior images with tested error rates [9, 10].

Practice thus offers two options. *Raw recording* keeps evidence of damage but exposes every identifiable bystander. *Irreversible anonymization* protects bystanders but also destroys the evidence that justifies recording in the first place. This paper studies a third option: **seal, don't blur**. Identity regions are separated inside the camera pipeline and stored as exactly recoverable encrypted pixels under per-person keys; the owner sees an anonymized scene; and identity is released only for one incident, with the owner's key or a court order.

Research questions.

RQ1: How much owner-accessible identifiable exposure does sealing remove relative to the strongest realistic baseline, event-triggered raw recording?

RQ2: Does sealing retain the evidence utility that irreversible anonymization loses?

RQ3: How much do fail-closed sealing and per-person keys each contribute, and under which detector weaknesses does protection fail?

RQ4: How do lighting, weather, pedestrian density and parking layout change exposure and evidence?

Contributions.

1. A sealed-by-default architecture for parked-vehicle cameras combining in-pipeline identity separation, exact encrypted recovery, head-region fail-closed sealing, per-person keys and an owner-or-court unseal rule (Section V).

2. Four propositions, the central one bounding residual exposure by the person-detector miss rate at the identification resolution (Section IV).

3. A model-independent primary metric, *accessible identifiable exposure* (AIE), computed on simulator ground truth at the IEC 62676-4 identification level, and a CPU-only simulator, to be released openly, with ray-traced occlusion, baselines, ablations and stress tests (Sections III and VI).

4. A paired benchmark of 810 episodes with clustered and episode-level bootstrap intervals and Holm-corrected tests, sensitivity and mechanism-cost analyses, a real-image check of the key assumption, and an explicit account of what the evidence does not show (Sections VII and VIII).

5. An analysis of the design against EU law as of October 2026, eight proposed regulatory amendments, and concept designs for official recording zones and driver guidance when recording stops (Section VIII).

Fig. 1 gives an overview of the proposed system: what the vehicle records, what the owner sees, where identity is kept, and who may release it.

Prior work already provides reversible face de-identification, threshold access to surveillance evidence and private geofencing (Section II). We claim neither primitive as new; our contribution is their integration for parked vehicles under a stated governance rule, and its quantitative evaluation.

II Related Work

Search protocol. On 5 October 2026 we searched a web index covering arXiv, IACR ePrint, IEEE/ACM/Springer/Elsevier abstract pages, USPTO and Google Patents, with 2024–2026 in depth and older seminal work where it defines a line. Queries covered dashcam anonymization, reversible face de-identification, threshold access to surveillance video, geofence-based camera control, privacy-preserving geofencing, cooperative sensing by parked vehicles and bystander recording indicators. Each query returned about ten results, so the survey is a lower bound on prior art; IEEE Xplore, Scopus and Espacenet were not yet queried directly[RSV].

Vehicle and dashcam privacy. Privacy Dashcam stores encrypted video and enforces anonymization on an external system before the user can view it [11]. Blockchain-based sharing protects the *owner's* location rather than bystanders [12]. Real-time face and plate blurring has been demonstrated on edge servers at smart-city intersections, with reported failures in difficult cases [13]. A legal analysis concludes that GDPR applies to both manufacturer and owner for pre-installed vehicle cameras, including live processing [14].

Reversible de-identification. Key-conditioned face anonymization that is recoverable only with the correct key is a mature line, including UU-Net [15], IdentityMask [16], RiDDLE [17], iFADIT [18] and diffusion-based methods [19]; a survey covers the area [20]. These methods recover an *approximate* face by generation, which matters for evidential use (Section V).

Governed access to evidence. Threshold secret sharing lets a quorum of key holders release a secret [21], and secret sharing has been used for collective control over surveillance video [22]. Pre-capture privacy imaging avoids acquiring faces and plates at all [23]. Privacy-preserving geofencing tests region membership without revealing location or region [24, 25, 26].

Patents and products. Vehicle recorders that change recording by geofence [27], fleet cameras that switch off in restricted sites [28], cameras that obey external restriction signals [29], parked vehicles that monitor a car park on behalf of others [30] and group-selective anonymization of faces and plates [31] are patented. Edge anonymization SDKs are commercially available [32]. Bystander studies show that single recording LEDs are weak notices and that visibility trades off against disruption [33, 34]. European guidance favours local processing and short retention for video devices and connected vehicles [35][RSV].

Positioning. Table I compares the closest approaches by capability.

What is not novel. Reversible key-based de-identification, threshold release of video, pre-capture privacy, edge face and plate blurring, geofence-controlled recording and private region membership tests all exist. To our knowledge, no work found in this search combines exact encrypted recovery, fail-closed sealing, per-person keys and an owner-or-court rule for parked vehicles and evaluates them against a ground-truth exposure metric; this statement must be re-checked against IEEE Xplore, Scopus and Espacenet.

III Problem Formulation

A parked ego vehicle carries cameras $c \in C$ with centre o_c, focal length f_c and field of view. Time is discretized into steps $t = 1, \dots, T$ of length Δt. Each person j has a face point $h_j(t)$ and facing direction $u_j(t)$; each number plate k has centre $p_k(t)$ and normal $n_k(t)$.

Ground-truth identifiability. For camera c at distance $d_{jc}(t) = \|h_j(t) - o_c\|$, the effective pixel density at the face is

$$\rho_{jc}(t) = q\big(\kappa, d_{jc}(t)\big)\,\frac{f_c}{d_{jc}(t)}, \tag{1}$$

where $q(\kappa, d) \in (0,1]$ is an image-quality factor for condition κ (Section VI). Person j is identifiable at time t if some camera sees the face inside its field of view, facing it within $\theta_{\max}$, at least a fraction $v_{\min}$ unoccluded, and at or above the identification density ρ_{id}:

$$\begin{aligned} I_j(t) = \max_{c \in C} \mathbb{1}[\mathrm{FOV}_c]\,\mathbb{1}\Big[u_j^{\top}\tfrac{o_c - h_j}{\|o_c - h_j\|} > \cos\theta_{\max}\Big] \\ \cdot\,\mathbb{1}[v_{jc} \geq v_{\min}]\,\mathbb{1}[\rho_{jc} \geq \rho_{\mathrm{id}}]. \end{aligned} \tag{2}$$

We use $\rho_{\mathrm{id}} = 250$ px/m, the identification level of IEC 62676-4 [36], $\theta_{\max} = 70^\circ$ and $v_{\min} = 0.5$. Plate identifiability $I_k(t)$ is defined analogously with 200 px/m (assumed) and $\theta_{\max} = 60^\circ$. Occlusion fractions v_{jc} are obtained by ray tracing.

Recording and protection. An event trigger produces a recording mask $r(t) \in \{0,1\}$. A design D applies a protection indicator $\pi_j^D(t) \in \{0,1\}$ to each person, computed only from the perception outputs: face detection $\delta_j^f(t)$ and person detection $\delta_j^p(t)$. Raw recording uses $\pi = 0$; anonymization and sealing use $\pi = \delta^f$, and their fail-closed variants use $\pi = \delta^f \vee \delta^p$. Plates are protected analogously by plate detection and, fail-closed, by vehicle detection.

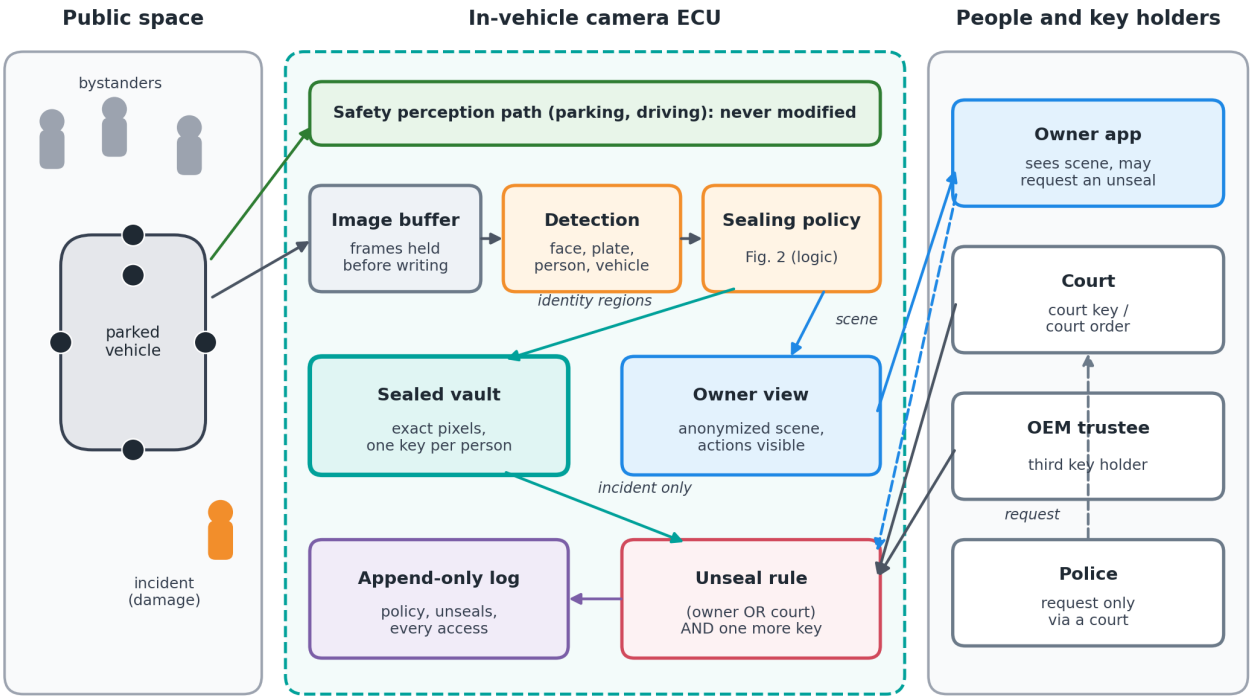


Figure 1: System architecture. The parked vehicle's cameras feed an in-vehicle image buffer; after detection, the sealing policy (Fig. 2) sends identity regions to a sealed vault (exact pixels, one key per person) and the anonymized scene to the owner's view. Identity leaves the vault only for one incident, under the rule (owner key OR court order) AND one further key holder; every decision and access is logged. The safety perception path for parking and driving is never modified.

Table I: Positioning of this work against the closest prior approaches.

Capability	Raw sentry	Irreversible anon. [9, 32]	Privacy Dashcam [11]	Reversible de-ID [15, 16, 17, 18, 19]	Threshold key sharing [21, 22]	This work
Identity hidden from owner view	No	Yes	Yes	Yes	No	Yes
Evidence recoverable	Yes	No	Yes	Approximate	Yes	Exact
Fail-closed on uncertain detections	–	Varies	No	No	–	Yes
Per-person release	–	–	No	Per face key	No	Yes
Owner-or-court rule	No	–	No	No	Quorum	Yes
Ground-truth exposure metric	No	Error rate	No	Re-ID scores	No	AIE

Accessible identifiable exposure (primary metric).

$$\mathrm{AIE}^D = \Delta t \sum_{t=1}^{T} r(t) \Big[\sum_j I_j(t) \big(1 - \pi_j^D(t)\big) + \sum_k I_k(t) \big(1 - \pi_k^D(t)\big) \Big]. \tag{3}$$

AIE counts identifiable person- and plate-seconds that a person can see without an authorized unseal. Because I is computed from geometry and π from the method's outputs, the metric never uses the method's internal estimate of whether someone is identifiable.

Evidence utility. For the person a who damages the vehicle, let $U_a^D(t) = I_a(t)$ for raw recording and for sealing (identity recoverable after an authorized unseal), and $U_a^D(t) = I_a(t)\big(1 - \pi_a^D(t)\big)$ for irreversible anonymization. Then

$$E^D = \mathbb{1}\Big[\Delta t \sum_t r(t)\, U_a^D(t) \geq \tau_E\Big], \qquad \tau_E = 1 \text{ s}. \tag{4}$$

Collateral exposure. An unseal for incident a over window W_a reveals other people's identity only if their regions share a key:

$$C^{\text{single}} = \Delta t \sum_{t \in W_a} r(t) \sum_{j \neq a} I_j(t), \qquad C^{\text{per-person}} = 0. \tag{5}$$

Owner-view utility. Let Φ_a be the recorded frames in which the body of person a is visible to some camera, and $w_a^D(t) = 1$ if design D seals a's whole body at t. The share of incident

Table II: Quantities of Section III and where they are measured.

Quantity	Answers	Measured in
AIE (3)	RQ1, RQ3, RQ4	Tables IV, V, VI, VIII; Figs. 3–5
E (4)	RQ2, RQ4	Tables IV, V, VIII, IX; Figs. 3, 4
C (5)	RQ3	Tables IV, V, IX; Fig. 4
O (6)	RQ3	Table IX
I_j, ρ (1), (2)	all	ground truth of every result; Table VII

frames in which the owner can still see what happened is

$$O^D = \frac{1}{|\Phi_a|} \sum_{t \in \Phi_a} \big(1 - w_a^D(t)\big). \tag{6}$$

The design goal is AIE as low as fail-closed anonymization, E as high as raw recording, $C = 0$, and O as high as raw recording. Table II maps each quantity to the results that measure it.

IV Theory

The following propositions use the definitions of Section III; Propositions 1 and 2 are elementary and stated for completeness, Proposition 3 is the main result.

Proposition 1 (No added exposure)**.** *For any design that stores identity only as sealed regions,* $\mathrm{AIE}^{\mathrm{seal}} \le \mathrm{AIE}^{\mathrm{raw}}$. *If sealing and irreversible anonymization use the same protection rule* π*, then* $\mathrm{AIE}^{\mathrm{seal}} = \mathrm{AIE}^{\mathrm{anon}}$.

Proof. Every summand of (3) is $r(t)I(t)(1-\pi(t))$ with $r, I, \pi \in \{0,1\}$, hence at most $r(t)I(t)$, the raw summand. AIE depends on the design only through π. □

Proposition 2 (Evidence preservation)**.** *If sealed regions are stored with lossless authenticated encryption, recording masks are equal, and the unseal rule is satisfied for incident a, then* $E^{\mathrm{seal}} = E^{\mathrm{raw}} \ge E^{\mathrm{anon}}$.

Proof. With exact recovery $U_a^{\mathrm{seal}} = I_a = U_a^{\mathrm{raw}}$; for anonymization $U_a^{\mathrm{anon}} = I_a(1-\pi_a) \le I_a$, so its sum and indicator cannot exceed the raw ones. □

Generative reversible methods [15, 16, 17, 18, 19] recover an approximation of the face; Proposition 2 then holds only up to reconstruction fidelity, which is why the architecture stores exact pixels and uses generation only for the owner's display.

Proposition 3 (Fail-closed bound)**.** *Assume (A1) whenever a face is identifiable in camera c, the same person's body lies at the same distance from c and is not more occluded than the face, so its effective density is at least* ρ_{id}*; and (A2) the person detector misses a person at density at least* ρ_{id} *with probability at most* ε*, conditionally on the face being identifiable. Then for fail-closed sealing*

$$\mathbb{E}\big[\mathrm{AIE}_{\mathrm{face}}^{\mathrm{seal,fc}}\big] \le \varepsilon\, \mathbb{E}\big[\mathrm{AIE}_{\mathrm{face}}^{\mathrm{raw}}\big]. \tag{7}$$

Proof. A face-second leaks only if $r(t) = 1$, $I_j(t) = 1$ and $\pi_j(t) = 0$, which requires $\delta_j^p(t) = 0$. By (A1) the body is at density at least ρ_{id}, so by (A2) $P(\delta_j^p(t) = 0 \mid r(t) = 1, I_j(t) = 1) \le \varepsilon$. Linearity of expectation gives (7). □

IEC 62676-4 itself places reliable person detection (25 px/m) an order of magnitude below identification (250 px/m) [36], which makes (A2) plausible by design of the standard. Proposition 3 makes the key assumption explicit and testable: fail-closed sealing protects as long as people are detectable at a lower pixel density than faces are identifiable. Section D stresses (A2) in simulation and Section E checks it on real images. Over 135 episodes, (A1) was violated in 1.2 % of identifiable face–camera steps overall and in up to 5.9 % at night (Section F).

Proposition 4 (Collateral exposure)**.** *With one key per incident clip, Poisson pedestrian arrivals of rate* λ *with independent trajectories, and ignoring occlusion between pedestrians,* $\mathbb{E}[C^{\mathrm{single}}]$ *is non-decreasing in* λ*. With per-person keys and correct track-to-key association,* $C^{\text{per-person}} = 0$.

Proof. For $\lambda' > \lambda$ a Poisson process of rate λ' is the superposition of one of rate λ and an independent one of rate $\lambda' - \lambda$; without mutual occlusion each added pedestrian contributes a non-negative term. With per-person keys, the unseal decrypts only regions keyed to a's tracks. □

Mutual occlusion can make extra pedestrians hide others, so Proposition 4 is not guaranteed in the simulator; Section C reports the measured trend and Section G the effect of tracking errors.

V Method

Fig. 1 shows the architecture and Fig. 2 its decision logic. The image pipeline separates identity regions from the scene before frames are written; the owner's view contains only the anonymized scene; identity regions are encrypted per person and released only under the unseal rule; and every policy decision and access is written to an append-only log. The privacy policy acts only on the recording and storage path; the perception path used by parking and driving functions is never modified, preserving freedom from interference in the sense of ISO 26262 [37].

Algorithm 1 lists the per-frame procedure shown in Fig. 2(a). **Unseal**$(a, \text{requesters})$ verifies that the requesters satisfy the rule, decrypts only the regions keyed to the tracks of incident a, and logs the request, the decision and every view. A manufacturer and police together cannot satisfy the rule, and requests spanning many vehicles require a court. The rule is a monotone, non-threshold access structure and can be realized with standard secret-sharing constructions [21]. Sealing only the head region, not the whole person, keeps the person's actions visible to the owner (Section G).

Complexity. Detection dominates and is $O(P)$ in pixels per frame. Sealing adds $O(N)$ region bookkeeping and authenticated encryption linear in the sealed pixel area A_s, so the

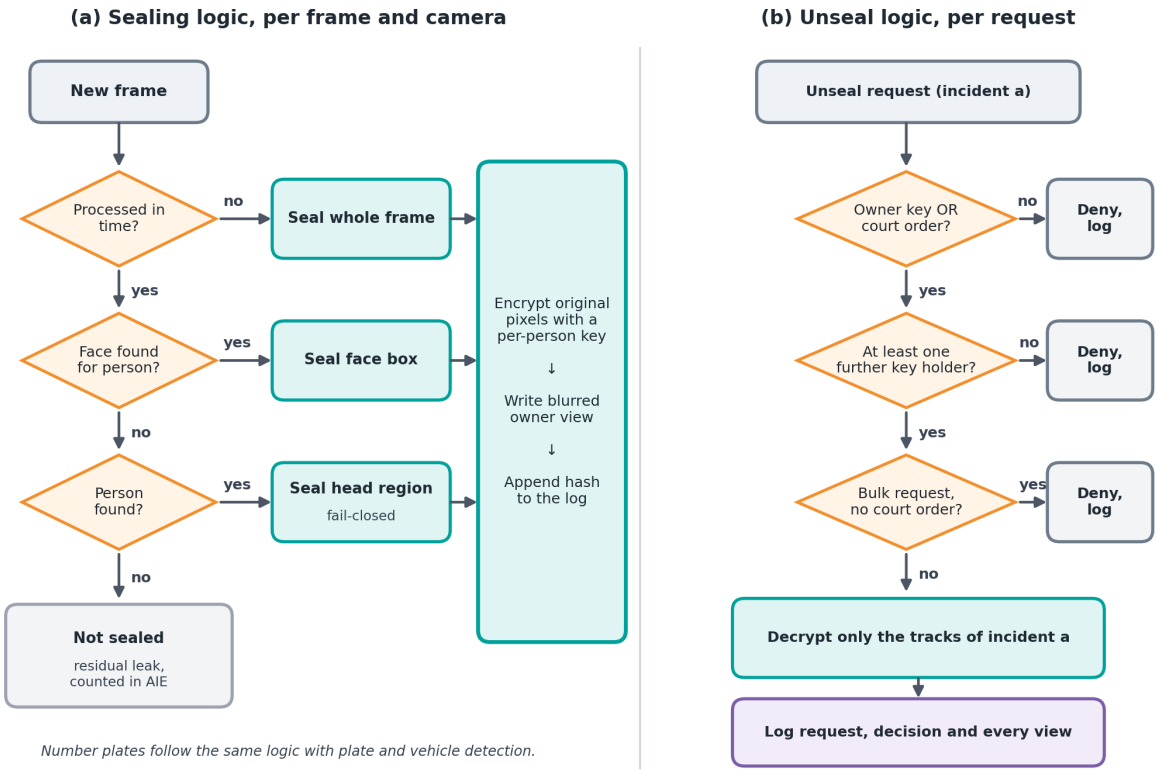


Figure 2: Decision logic. (a) Per frame and camera: late frames are sealed whole; a person's face box is sealed when found, otherwise the head region is sealed (fail-closed); only a person missed by both detectors remains unsealed, which is exactly what AIE counts. (b) Per unseal request: release requires the owner key or a court order plus a further key holder, bulk requests require a court, and only the incident's tracks are decrypted.

Algorithm 1 Sealed-by-default recording (per frame, per camera)

1: Run person/vehicle and face/plate detection on the frame in the image-pipeline buffer.
2: Sealing set ← all face and plate boxes ∪ head regions of person boxes without a face box ∪ plate zones of vehicle boxes without a plate box.
3: **if** the frame deadline is missed **then**
4: Seal the whole frame; never write an unprocessed frame.
5: **end if**
6: **for** each sealed region with track τ **do**
7: Encrypt its original pixels with authenticated encryption under key k_τ; bind frame, camera and track identifiers as associated data.
8: **end for**
9: Write the owner view with sealed regions blurred or inpainted.
10: Wrap each k_τ under (owner key OR court key) AND one further key holder.
11: Append a hash of frame metadata, policy state and wrapped keys to the log.

overhead is $O(N + A_s)$ per frame; storage grows by A_s per recorded frame plus one wrapped key per track. In the simulator, ground-truth evaluation costs $O(|C|NKB)$ ray–box tests per step for N people, K face sample points and B occluders; one ten-minute episode runs in about 2.5 s on one CPU core.

Out of scope. Official recording zones (Figs. 6 and 7), private zone lookup and cooperative recording among parked vehicles are planned extensions (Section VIII) and are not evaluated here.

VI Experimental Setup

Simulator. A 2.5D geometric simulator (Python, NumPy; sim v3) places the ego vehicle between two parked neighbours, optionally facing a row of five parked cars across the front aisle. Pedestrians arrive as a Poisson process on a front sidewalk and a rear aisle, about 8 % of them cutting between cars; vehicles pass in both aisles. Each episode contains one incident: a person approaches, stands 0.35–0.9 m from a randomly chosen point along the vehicle for 6–20 s with face yaw up to ±60°, and leaves. Face heights vary per person (1.45–1.85 m). Occlusion is ray-traced from each camera to five points on each face, one point per plate and three points on each body, against vehicle and pedestrian boxes.

Cameras. Four equidistant fisheyes (1920 px across 190°) and a front pinhole camera (1920 px across 60°); the identification ranges at 250 px/m are 2.32 m and 6.65 m in clear daylight.

Conditions. Nine conditions combine day, dusk and night with clear, rain and fog. Time of day and rain scale effective density by fixed factors; fog uses Koschmieder contrast transmission $T(d) = \exp(-3.912\,d/V)$ [38] with assumed visibility $V = 100$ m. Morning and afternoon are not distinguished because sun direction and glare are not modelled. All factors are assumptions (Table III).

Perception model. Face, plate, person and vehicle detection probabilities are logistic in effective density, with condition shifts, a yaw penalty beyond 40°, an occlusion penalty, a per-track latent offset (correlated misses) and per-frame noise. The perception model is independent of ground truth by construction; a sanity check confirms that ground truth is unchanged when the perception model changes.

Designs. B0 continuous raw recording (context only); B1 event-triggered raw recording (10 s before, 20 s after a person stays within 1 m for 1 s), the strongest realistic baseline, with trigger and clip parameters assumed from publicly described sentry behaviour rather than a documented product configuration; B2 event-triggered irreversible blurring; B2fc the same with fail-closed protection, i.e., B2 in its best configuration; P the proposed design; and three ablations: A1 without fail-closed, A2 continuous sealed recording, A3 one key per clip.

Scenario catalogue and protocol. 9 conditions × 3 densities (2, 6, 15 pedestrians/min) × 2 layouts × 15 seeds = 810 ten-minute episodes; all designs run on identical scenes and perception outputs, giving paired observations. Stress tests vary the face-detector 50 % point (110–320 px/m) at two levels of correlated misses (240 episodes), the person-detector 50 % point (20–400 px/m; 120 episodes), and both jointly (80 episodes).

Statistics. Means with percentile bootstrap 95 % confidence intervals (2,000 resamples) [39]; paired Wilcoxon signed-rank tests for exposure [40] with rank-biserial effect sizes; exact McNemar tests for binary evidence [41]; Holm correction across the six primary tests [42]. Because the 810 episodes share 15 scene seeds across conditions, densities and layouts, we also report a cluster bootstrap that resamples seeds (4,000 resamples) and seed-level Wilcoxon tests on the 15 seed means, Holm-corrected. Ten sanity checks (projection, field of view, analytic identification range, occlusion, single-pedestrian onset, full occlusion, perfect and blind detectors, determinism, ground-truth independence) pass before every run; fixed seeds make all runs deterministic.

VII Results

All results are simulated (sim v3). Two earlier runs were discarded: v1 because its incident geometry was nearly fixed and its fail-closed path was never stressed, and v2 because fog was modelled independently of distance; non-fog episodes are numerically identical in v2 and v3.

Table III: Main parameters (assumptions unless cited).

Parameter	Value
Step length; episode	0.5 s; 600 s
Identification density, face; plate	250 px/m [36]; 200 px/m
Max facing angle, face; plate	70°; 60°
Min visible fraction	0.5
Time factor day / dusk / night	1.0 / 0.8 / 0.5
Weather factor clear / rain	1.0 / 0.8
Fog visibility; detector fog penalty	100 m; $-3.0\,(1 - T(d))$ logit
Face / plate / person / vehicle detector 50 % point	110 / 90 / 20 / 10 px/m
Correlated-miss σ; frame noise σ	0.6; 0.4 logit
Yaw penalty	2.0 logit per 30° beyond 40°
Trigger distance; hold; pre; post	1.0 m; 1.0 s; 10 s; 20 s
Evidence threshold τ_E	1.0 s

A Privacy and evidence by design (RQ1, RQ2)

Table IV summarizes all designs over 810 paired episodes; event-triggered designs record 31.1 % of the time. Table V reports the paired tests. Sealing removed 59.5 s of accessible exposure relative to B1 and added 0.52 evidence utility relative to B2fc (418 incidents recoverable versus none). As Propositions 1 and 2 predict, it equalled B2fc in exposure and B1 in evidence in every episode. These two ties hold by construction: the proposed design's exposure and evidence columns quantify the trade-off between the baselines rather than test the proposal. The method-dependent evidence is in the stress tests (Section D), the per-person-key ablation (Section C), the mechanism costs (Section G) and the real-image check (Section E). Rank-biserial values of ± 1 follow from dominance by construction, so p-values are trivially small; confidence intervals carry the information. Treating seeds as clusters gives -59.5 s $[-64.0, -55.4]$ for T1, $+0.52$ $[0.45, 0.58]$ for T2 and $+1.08$ s $[0.88, 1.30]$ for T6, with seed-level Holm-corrected $p = 0.0013$, 0.0013 and 0.0002 ($n = 15$).

B Conditions and layout (RQ4)

Fig. 3 shows exposure and evidence by condition. Evidence utility, identical for B1 and P, was 1.00 in day clear, 0.93 in day fog, 0.67 in day rain and dusk clear, 0.62 in dusk fog and dusk rain, 0.13 at night in clear weather, and 0 at night in rain or fog; B2fc was 0 everywhere. Night evidence is limited by optics, not by the privacy design. Day fog scores above day rain because fog attenuates little within identification range while rain is modelled as distance-independent. Plate exposure was 98.0 s with a facing row of parked cars and 0 without; face exposure was 10.5 s in both layouts.

C Density and per-person keys (RQ3, RQ4)

Fig. 4 shows the density trend for exposure, evidence and collateral exposure. Raw face exposure rose from 3.4 to 7.4 to 20.7 s at 2, 6 and 15 pedestrians/min. Collateral exposure with one key per clip rose from 0.44 to 0.71 to 2.09 s, consistent with Proposition 4; per-person keys kept it at 0.

Table IV: Mean per ten-minute episode with bootstrap 95 % CIs (810 paired episodes, sim v3).

Design	AIE (s) [95 % CI]	of which faces (s)	Evidence utility [95 % CI]	Collateral per unseal (s)
B0 continuous raw	178.1 [149.0, 208.2]	18.1	0.52	–
B1 event raw	59.5 [49.5, 69.6]	10.5 [9.7, 11.3]	0.52 [0.48, 0.55]	–
B2 blur	0.20 [0.15, 0.25]	0.04	0.00	–
B2fc blur, fail-closed	0.00	0.00	0.00	–
P proposed	**0.00**	0.00	**0.52 [0.49, 0.55]**	0.00
A1 no fail-closed	0.20 [0.15, 0.25]	0.04	0.52	0.00
A2 continuous sealed	0.00	0.00	0.52	0.00
A3 one key per clip	0.00	0.00	0.52	1.08 [0.98, 1.18]

Table V: Paired tests over 810 episodes, Holm-corrected.

Test	Mean difference [95 % CI]	Effect	p
T1 AIE, P − B1	−59.5 s [−70.0, −50.4]	$r_{rb} = -1.0$ (776 non-zero)	< 0.001
T2 Evidence, P − B2fc	+0.52 [0.48, 0.55]	418 vs 0 discordant	< 0.001
T3 AIE, P − B2fc	0 (all pairs equal)	–	1.0
T4 Evidence, P − B1	0 (no discordant pairs)	–	1.0
T5 AIE, A1 − P	+0.20 s [0.15, 0.25]	$r_{rb} = +1.0$ (127 non-zero)	< 0.001
T6 Collateral, A3 − P	+1.08 s [0.98, 1.18]	$r_{rb} = +1.0$ (504 non-zero)	< 0.001

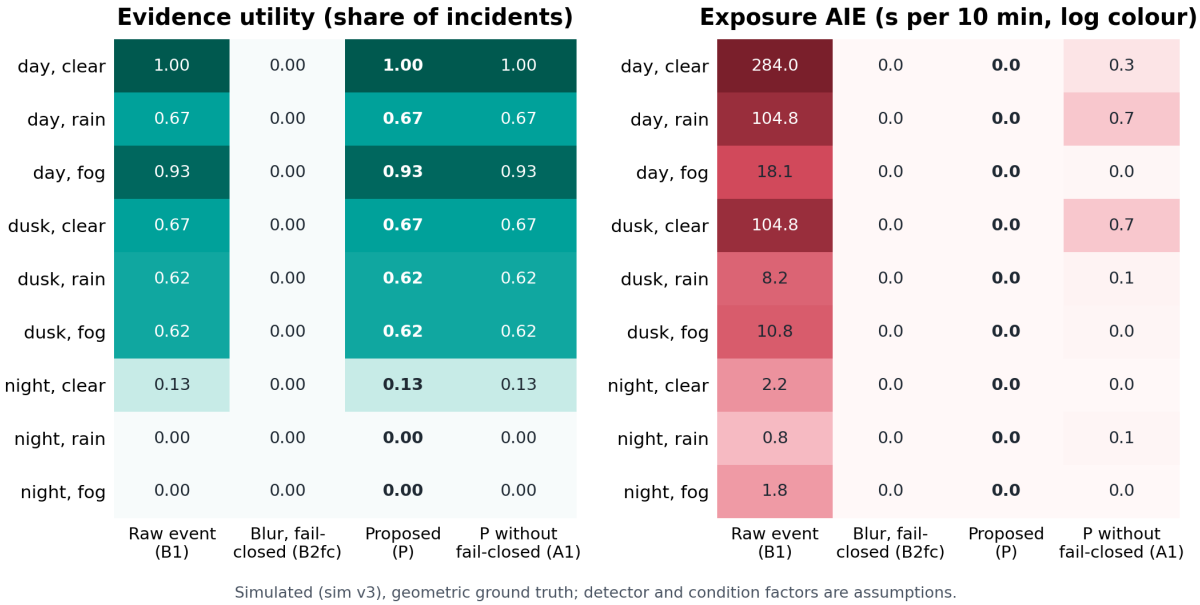


Figure 3: Evidence utility (left, teal scale) and accessible identifiable exposure in s per ten minutes (right, red logarithmic scale) by condition and design; the proposed design is the bold third column. Simulated, geometric ground truth; detector and condition factors are assumptions.

D Stress tests (RQ3)

Fig. 5 varies the face detector. Without fail-closed (A1), day face exposure rose from 0.07 to 0.4, 2.1 and 6.6 s as the face detector's 50 % point moved from 110 to 320 px/m (raw: 15.5 s); P stayed at 0. Weakening only the person detector changed nothing, because the face detector already covered nearly all identifiable faces. Table VI combines a weak face detector (320 px/m) with a progressively weaker person detector. Fail-closed protection held until person detection required the identification density itself, as Proposition 3 predicts. When blurring fails in this regime its evidence utility rises (0.9 at 400 px/m) because leaked faces become usable: a privacy failure that looks like an evidence success.

Table VI: Joint stress, day clear: face-seconds visible per ten minutes with a weak face detector (320 px/m).

Person detector 50 % point	B1 raw	A1	P
20 px/m	16.4	6.6	0.00
120 px/m	16.4	6.6	0.00
250 px/m	16.4	6.6	0.05
400 px/m	16.4	6.6	5.50

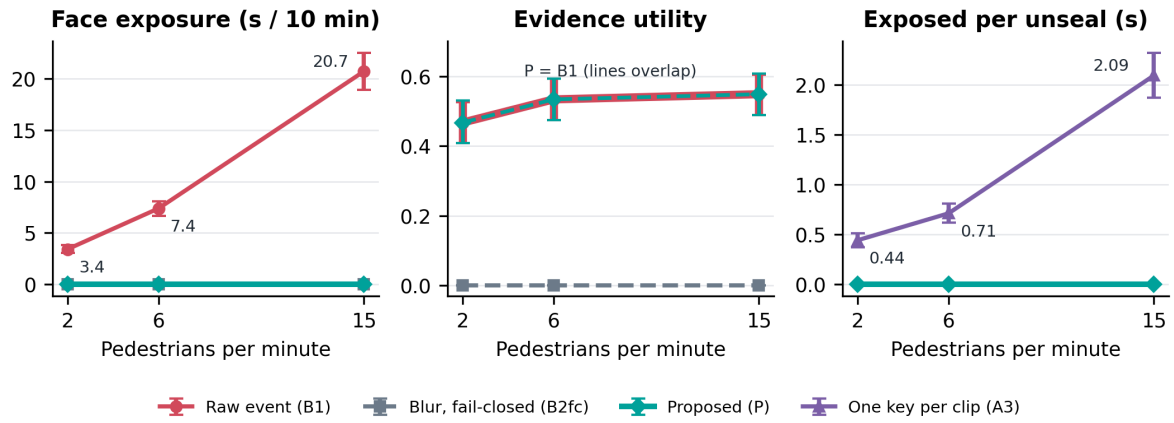


Figure 4: Versus pedestrian density: face exposure (left), evidence utility (middle; proposed and raw recording overlap exactly) and bystander exposure revealed by one unseal (right; one key per clip versus per-person keys). Means with bootstrap 95 % CIs; labels give the values quoted in Section C.

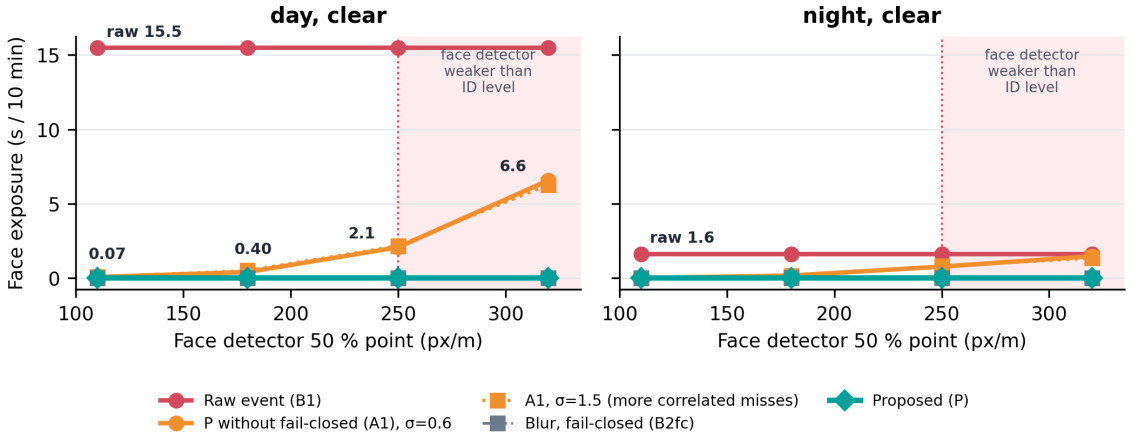


Figure 5: Face exposure versus the face detector's 50 % point, day and night clear, for two levels of correlated misses (σ). The shaded region marks detectors weaker than the identification level (250 px/m); orange labels give the A1 values quoted in Section D; the proposed design stays at 0 throughout.

E Real-image check of assumption (A2)

Proposition 3 rests on people being detectable at lower pixel density than faces are identifiable. We checked this on four phone photographs (24 mm-equivalent main camera, civil dusk, ISO 640–1,000) of one adult standing beside parked cars (consent required before publication[RSV]). Each photo was downscaled so that the face covered a target density, assuming a 0.20 m face box, and classical detectors were run: a Viola–Jones face cascade [43] and a HOG person detector [44], natively and on 2× upsampled images. A detection counted only if it covered the subject. Table VII shows the person detectable at roughly 4–10× lower density than the 250 px/m identification level, consistent with (A2). This check involves one subject and a phone camera whose night processing exceeds automotive cameras; it says nothing about fisheye distortion, sensor noise, motion blur, night, rain, crowds or modern automotive detectors, and downscaling emulates resolution only. Viola–Jones and HOG are older detectors; modern detectors would likely detect at lower density, making these limits conservative for (A2), but this was not tested.

Table VII: Lowest density at which each detector found the subject.

Photo	Face	Person	Person, 2× up
Frontal 1	110	60	32
Frontal 2	110	50	25
Frontal 3	160	60	32
Profile	excluded (reference face detection was a false positive)		

All values in px/m.

F Sensitivity analysis

Table VIII varies the parameters a reviewer is most likely to question, on 135 episodes (nine conditions, 6 pedestrians/min, facing row of cars, 15 seeds) plus 90 episodes for condition factors. The proposed design's exposure stayed at 0 and its

evidence equalled raw recording's in every variant, and fail-closed blurring's evidence stayed at 0; absolute magnitudes, however, moved by more than an order of magnitude. Raw-recording exposure in this subset (98.0 s at baseline) exceeds the Table IV mean (59.5 s) because every subset episode has a facing row of parked cars, whose plates dominate exposure. A 0.5 m trigger misses most incidents (evidence 0.17), so trigger design affects evidence more than privacy design does. The rain factor crosses the plate-identification threshold of the facing row of cars between 0.7 and 0.9, which explains the jump in exposure. Assumption (A1) was violated in 1.2 % of identifiable face–camera steps (0.7–0.9 % in daylight, up to 5.9 % at night).

G Costs and robustness of the mechanism

Three mechanisms the headline results take for granted were simulated on the same 135 episodes (Table IX); owner-view utility is O^D from (6). The protection rule $\pi = \delta^f \vee \delta^p$, and therefore every exposure result above, does not depend on how large the sealed region is. Whole-person fail-closed sealing would hide the person's actions in about four of five incident frames, because people facing the car rarely show a detectable face. Sealing only the head region removes this cost at equal protection under assumption (A3): the head region estimated from a person box covers the face; (A3) is not tested here. Per-person keys stayed robust to tracking errors, and sealing late frames whole, as Algorithm 1 prescribes, preserved evidence at every miss rate, whereas discarding them cost up to 0.12 in evidence utility.

VIII Discussion

What the results mean. Sealing does not beat fail-closed blurring on privacy or raw recording on evidence; it obtains both at once. The practical claim is therefore narrow: if a vehicle records bystanders at all, sealed, fail-closed recording removes the trade-off that today forces a choice between protecting bystanders and protecting the owner.

HMI. The owner sees an anonymized scene and must understand that identity exists but is locked. Wording such as "faces sealed; unseal needs your key or a court order" and an explicit unseal request flow are needed to avoid the impression of a broken camera; head-only sealing lets the owner see what happened without seeing who did it. Bystanders need a notice they can act on; single lights are weak notices [33, 34], so an external display with a link to a rights page is a candidate for a user study.

Functional safety and SOTIF. The policy acts only on the recording and storage path, so the perception path of parking and driving functions is unchanged (ISO 26262 [37]). Sentry recording is not a safety function, but driving-mode recording may contain safety-relevant data; privacy rules should apply only to optional camera recording, and the boundary with event data recording needs a SOTIF analysis (ISO 21448 [45]). Key storage and the unseal service fall under vehicle cybersecurity management (UN R155 [46]).

Operational design domain. Results hold for a parked vehicle in a surface car park with the modelled camera set. Multi-storey and underground car parks, on-street parking, crowds above 15 pedestrians/min and other camera configurations are outside the evaluated domain.

Regulation. The design addresses the controller problem identified by regulators [1, 3, 4] by keeping identity out of the owner's view, but it does not by itself make recording lawful. Sealed regions are pseudonymized, not anonymized, personal data in the sense of GDPR Art. 4(5): the owner remains controller, and a lawful basis, retention limits and a data protection impact assessment are still required. The EU driver-monitoring requirement shows that purpose-bound, on-board camera rules are feasible [47]; the EU Data Act may increase third-party access to vehicle data, which strengthens the case for sealing at the source [48]. German case law accepts event-based clips as evidence while rejecting permanent recording [49]. A court key infrastructure does not exist and would require legislation.

Threats to validity. *Construct:* AIE uses a geometric identifiability proxy at a standardized density rather than a face-recognition model on rendered images; identification thresholds, detector curves and condition factors are assumptions. *Internal:* sealing and fail-closed blurring share the same perception outputs, so their exact tie is by construction; glare is not modelled, and tracking errors and frame-deadline misses are modelled only in the mechanism study. *External:* one parking geometry family, one camera set and one incident type. *Conclusion:* paired tests on deterministic dominance produce extreme p-values; we rely on confidence intervals, effect sizes and seed-clustered intervals.

Limitations. No real vehicle, real automotive camera or real bystander data were used; the real-image check involves one subject and a phone camera. Night evidence is near zero for every design. Context such as clothing can identify people even when faces are sealed. Only one incident type (a person standing beside the vehicle) is simulated, and head-region sealing rests on the untested assumption (A3).

Validation plan. A capture campaign with a production surround-view set and a reference camera on a test vehicle, calibrated and time-synchronized, with 10–20 consenting participants, ground distance markers, yaw from 0° to 90°, day/dusk/night and dummy plates on private ground, will measure person- and face-detection limits, head-region coverage and identifiability against density per camera and condition. The decisive test is whether the person-detection limit stays below the identification limit everywhere; where it does not, fail-closed sealing fails.

Extension: official recording zones (concept, not evaluated). Some places need stricter rules than an ordinary street. We propose that only competent authorities declare such zones, through an official sign at the entrance and a digitally signed register entry distributed with map data (Fig. 6). The vehicle learns a zone before it arrives through a private tile query, so the register never reveals the list of sensitive sites [24, 25, 26], and confirms it on arrival by reading the sign; a sign without a register entry is ignored and reported.

Table VIII: Sensitivity of raw-recording exposure and evidence utility (proposed = raw for evidence; proposed exposure 0 throughout).

Parameter varied	Values	Raw AIE (s / 10 min)	Evidence utility
Evidence threshold τ_E	0.5 / 1 / 2 / 3 s	unchanged	0.59 / 0.53 / 0.41 / 0.31
Clip length (pre/post)	5/10, 10/20, 20/40 s	62.0 / 98.0 / 156.7	0.53 for all
Trigger distance	0.5 / 1.0 / 1.5 m	7.5 / 98.0 / 124.6	0.17 / 0.53 / 0.53
Night factor (night, clear)	0.35 / 0.5 / 0.65	0.0 / 1.6 / 6.3	0.00 / 0.07 / 0.67
Rain factor (day, rain)	0.7 / 0.8 / 0.9	8.1 / 175.2 / 499.7	0.67 / 0.73 / 0.93

Table IX: Mechanism costs and robustness, means (bootstrap 95 % CIs) over 135 episodes.

Mechanism	Setting	Result
Owner view of the incident (share of recorded incident frames with the person's actions visible)	Raw recording	1.00
	Fail-closed, whole person sealed	0.21 [0.18, 0.23]; night 0.13
	Fail-closed, head region sealed	1.00 (assumes the head region covers the face)
Tracking ID swaps near the vandal (probability per step within 2 m)	0 / 0.01 / 0.05 / 0.1 / 0.2	collateral 0.00 / 0.00 / 0.00 / 0.02 / 0.03 s; evidence 0.53 $\rightarrow$ 0.51
Frame-deadline misses, frame sealed whole	0–50 %	evidence 0.53 at every rate
Frame-deadline misses, frame discarded	0 / 5 / 20 / 50 %	evidence 0.53 / 0.53 / 0.51 / 0.41

Zones only tighten the evaluated default (Fig. 7): Z0 streets keep the design of this paper; Z1 clinics or hospitals and places of worship mask entrances and allow unsealing only by court; Z2 schools protect minors with court-only unsealing; and Z3 sites designated by statute, such as military or critical infrastructure, stop recording, storage and remote viewing entirely, which also forgoes evidence there. Where satellite positioning fails, as in underground car parks, the vehicle keeps the class of its last verified entry, and on conflicting or uncertain inputs the stricter class wins. None of this was simulated: zone classes change unseal rights, retention and capture, which the present metrics and scenarios do not yet cover.

Driver guidance when recording stops (concept, not evaluated). Stopping recording must never stop safety. Camera-based functions such as emergency braking and lane keeping are mandatory for new vehicles in the EU [47], and the privacy policy acts only on recording, storage and remote viewing, never on the perception path (Fig. 1). In a Z3 zone the cameras therefore keep feeding driver assistance with frames that are processed in volatile memory and discarded; what stops is every path by which images could be kept or seen. Fig. 8 shows the corresponding HMI. Before entry the driver is told that recording will stop while assistance stays active; while driving, a recording-off indicator appears next to an assistance-on indicator; when parked, camera-based sentry is unavailable and the alarm falls back to the non-imaging sensors of a conventional vehicle alarm (motion, tilt, ultrasonic, radar) [50], notifying the owner without images. If a site's statute forbade operating cameras at all, the vehicle would enter the same degraded mode as for a blocked or soiled camera, a case that functional-safety and SOTIF analyses already have to cover [37, 45]: camera-dependent functions are announced as unavailable, parking falls back to ultrasonic guidance, and the driver is asked to take extra care. Such a ban would conflict with mandatory camera-based safety functions; resolving that conflict is for regulators, not for the vehicle.

Further extension. Cooperative recording among parked vehicles, in which co-located cars agree which of them records, is designed but not evaluated here.

Regulatory alignment and proposed amendments. Table X relates the design to the instruments in force or under negotiation in October 2026 and lists the amendments it suggests; these are policy proposals derived from the design, not legal advice. The European Data Protection Board's 2025 guidelines confirm that pseudonymised data remain personal data while pseudonymisation supports legitimate interest and data protection by design [51]; the Court of Justice has since held that strongly pseudonymised data need not be personal for a recipient unable to re-identify [52], which would cover a backend that stores sealed regions without keys, but leaves open whether a holder of one key share can re-identify. The proposed codification of that relative concept in the GDPR remains contested [53]; the design does not depend on its outcome. The Data Act has applied since September 2025, with design obligations for new products from September 2026 [48], so sealed identity regions should be excluded from user and third-party access. Face and person detection used only for sealing does not identify anyone, and the amended AI Act postpones high-risk obligations to 2027–2028 [54]. Official recording zones would require a road-data category for authority-issued zone records [55].

IX Conclusion

Parked-vehicle cameras today force a choice between exposing bystanders and destroying evidence. We formulated that choice with four ground-truth quantities—accessible identifiable exposure, evidence utility, collateral exposure and owner-view utility—and proposed sealing identity at the source, failing closed on uncertain detections, keying each person separately and requiring the owner or a court to unseal. In simulation all four targets were met together: across 810 paired episodes, exposure fell from 59.5 s to 0.0 s per ten minutes rel-

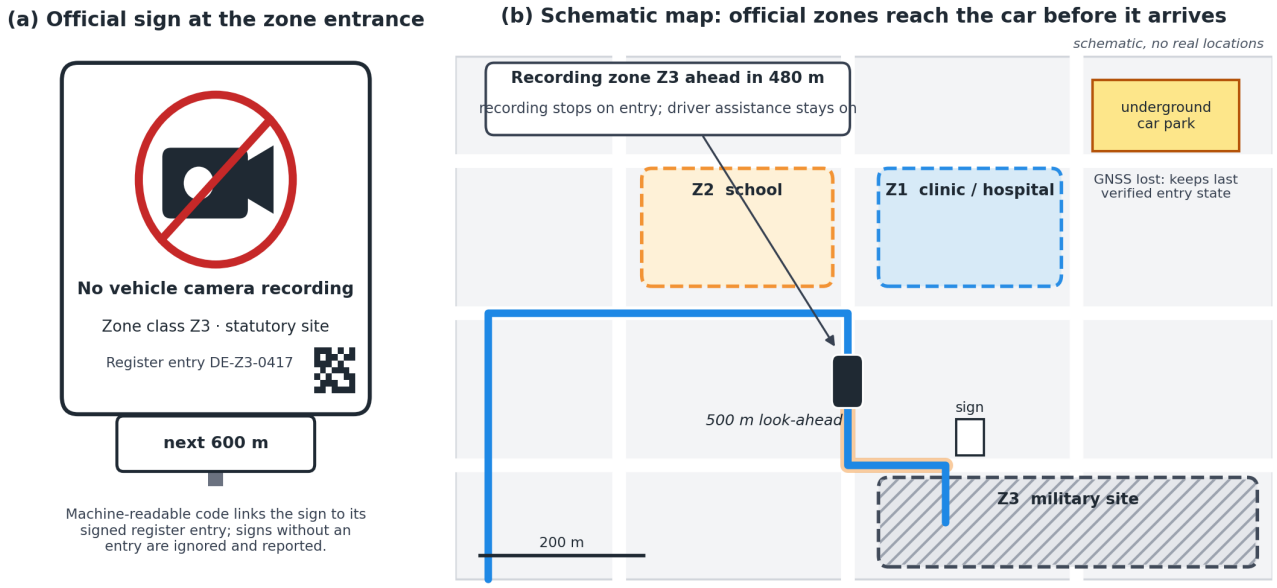


Figure 6: Official recording zones, concept (not evaluated). (a) Official sign at a zone entrance, linked by a machine-readable code to its signed register entry. (b) Schematic map (no real locations): the signed zone record reaches the vehicle through the map look-ahead before arrival; the sign confirms it on entry; in a car park without satellite positioning the last verified entry state is kept.

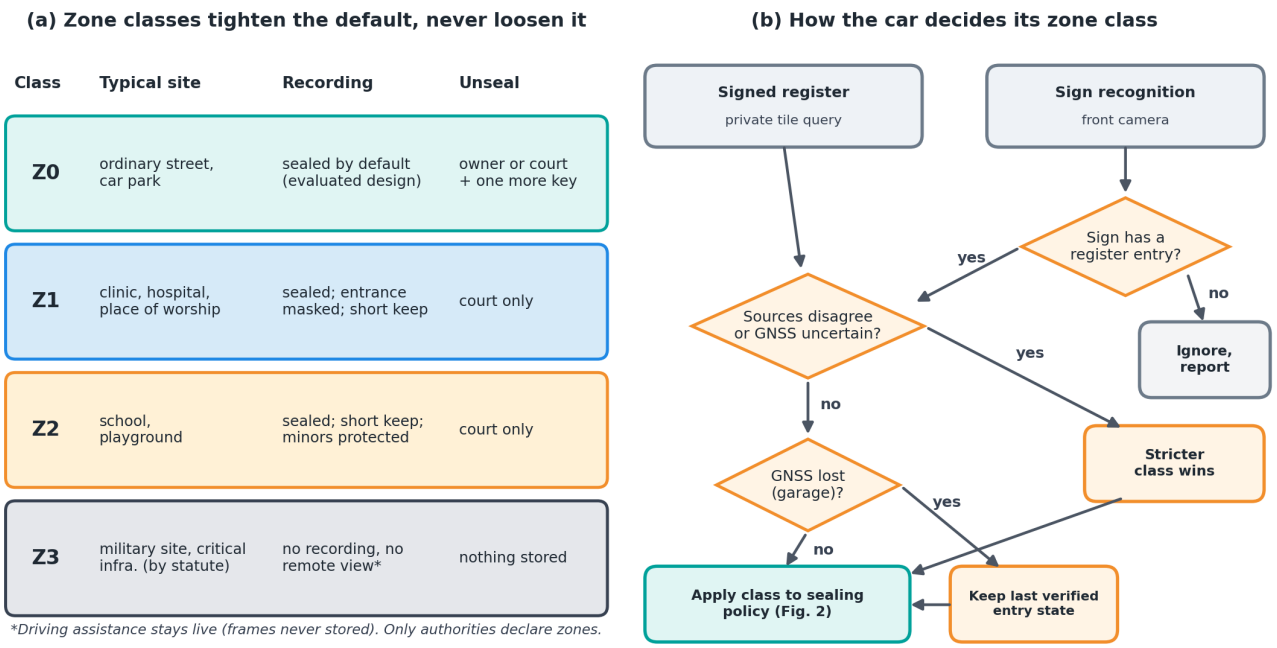


Figure 7: Zone classes and zone decision logic, concept (not evaluated). (a) Classes tighten the evaluated default; only Z3 sites designated by statute stop recording, while live perception for driving continues. (b) The register is the source of truth: a sign without a register entry is ignored, the stricter class wins when sources disagree or positioning is uncertain, and the last verified entry state is kept when positioning is lost. The resulting class parameterizes the sealing policy of Fig. 2.

ative to event-triggered raw recording; evidence utility stayed at 0.52 instead of falling to zero as with blurring; per-person keys avoided 1.08 s of bystander exposure per unseal; and head-region sealing kept the incident visible to the owner in

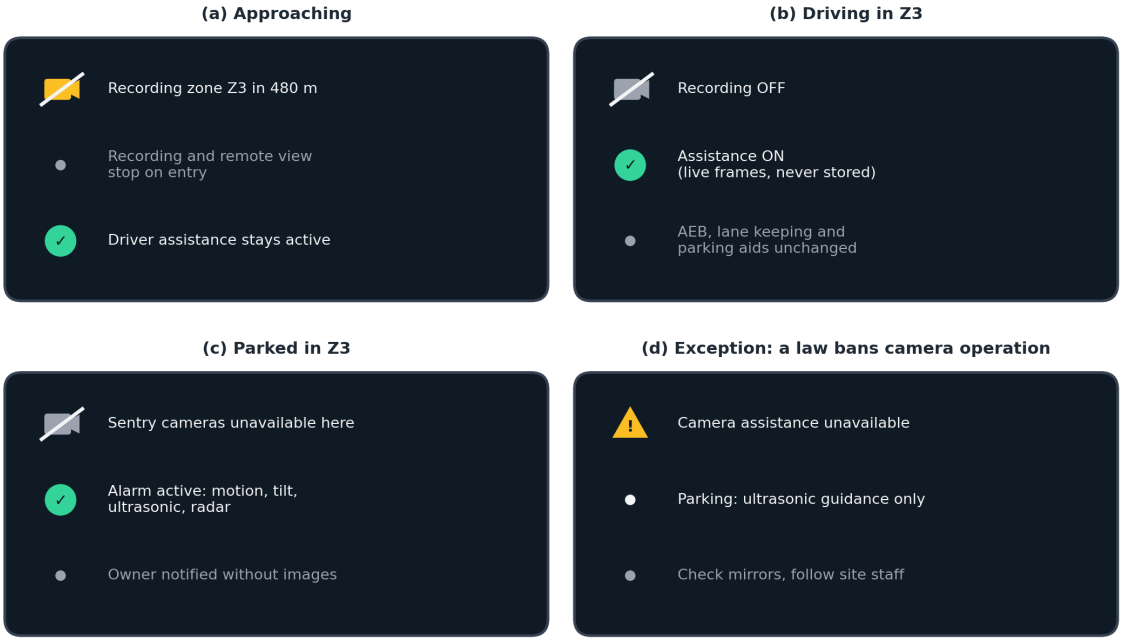


Figure 8: Driver guidance in a Z3 zone, HMI concept (not evaluated). (a) Before entry: recording and remote view will stop, assistance stays active. (b) Driving: recording off, assistance on with live frames that are never stored. (c) Parked: camera-based sentry unavailable; the alarm uses non-imaging sensors and notifies the owner without images. (d) Exception where a law bans camera operation: camera assistance is announced as unavailable and parking falls back to ultrasonic guidance.

Table X: Regulatory alignment (status October 2026) and amendments proposed by this work. Proposals are derived from the design and are not legal advice.

Instrument (status)	How the design aligns	Remaining gap	Proposed amendment
GDPR; EDPB Guidelines 01/2025 [51]	Sealing is pseudonymisation with keys held apart; strengthens the legitimate-interest balancing	Data protection by design binds the owner, who cannot design the system	P1: manufacturer duty at type approval for on-board sealing, event triggers and retention limits
EDPS v SRB, 2025 [52]	A backend holding sealed regions without keys cannot re-identify	Status of a holder of one key share is unclear	P2: guidance recognising split-key sealing as effective pseudonymisation
Data Act [48] (2025; design duties 2026)	Owner view contains no identity	Sealed regions could be requested as product data	P3: exclude sealed regions from Data Act access
GDPR procedure and national law [2, 3, 49]	Owner-or-court rule, incident-scoped release, logging	No legal basis for a court key holder	P4: judicial unseal procedure; bulk requests only by court order
General Safety Regulation [47]	Privacy policy never touches the safety path	Camera bans could collide with mandatory safety functions	P5: separate recording from perception in law
Road traffic and ITS law [55]	Zones only tighten the default	No authority-issued recording zones	P6: official sign plus signed register entry; zone classes Z0–Z3
Type-approval testing	AIE measures bystander exposure on ground truth	No measurable privacy requirement	P7: standard exposure test, comparable to GB/T 44464 [9]
UN R155/R156 [46]	Vault and unseal service under cybersecurity management	Recording features added by software update	P8: privacy assessment for updates that change recording
AI Act, amended 2026 [54]	Detection for sealing, no identification	None identified; classification to confirm	–

every frame, against 21 % with whole-person sealing. These magnitudes depend on assumed detector, trigger and optics models, while the relative conclusions held in every sensitivity run. The protection rests on people being detectable at lower resolution than faces are identifiable and on head regions covering faces—assumptions supported by the DORI levels of IEC 62676-4 and weakly by three real images, and still to be tested with automotive cameras. Beyond the design, the analysis identifies where EU law should change: a manufacturer duty for privacy by design of parked recording, recognition of split-key sealing as pseudonymisation, exclusion of sealed regions from Data Act access, and a judicial unseal procedure. Real-vehicle capture, official zones and cooperative recording are the next steps.

Declarations

Data and code availability. Simulator, sanity checks, benchmark scripts, per-episode results, analysis and video build scripts are archived on Zenodo at `https://doi.org/10.5281/zenodo.23575914` (code MIT, data CC BY 4.0) and reproduce every simulated value in this paper [56]. A 108-second explainer video (1080p, with subtitles) is provided as supplementary multimedia; its results are the simulated values reported here. Personal photographs are not released. **Conflict of interest.** This work is an independent personal research project. The author is employed in the automotive industry; the views and the regulatory proposals in Section VIII are the author's own and do not represent the position of any employer. **Funding.** None declared (to confirm). **Image consent.** Photographs used in Section E show an identifiable person; written consent for the analysis is required[RSV]. No images of people are reproduced in this paper.

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